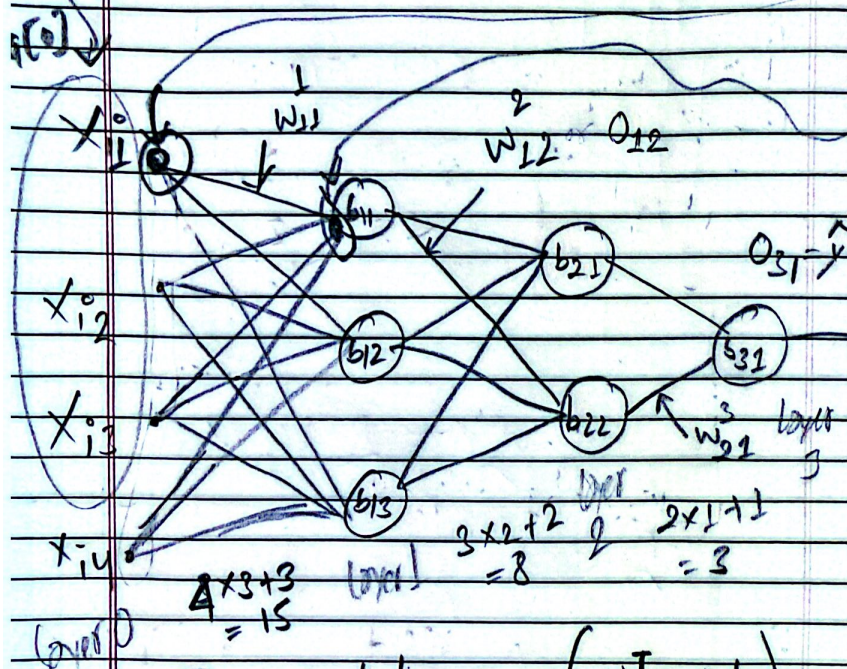


forward propagation $\rightarrow$ no learning happens

Input

cgpa	iq	10 th marks	12 th marks	placed
7.2	72	65	81	1
8.1	92	75	76	0



Output $\rightarrow$ Prediction $\rightarrow \sigma(W^T X + b)$
 Note: form $\rightarrow$ faster for computer!
 Total trainable parameters = $15 + 8 + 3 = 26$

$$\text{Total weights} = 4 \times 3 + 3 \times 2 + 2 \times 1 = 12 + 6 + 2 = 20$$

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Layer # 1

$$\begin{bmatrix} w_{11}^1 & w_{12}^1 & w_{13}^1 \\ w_{21}^1 & w_{22}^1 & w_{23}^1 \\ w_{31}^1 & w_{32}^1 & w_{33}^1 \\ w_{41}^1 & w_{42}^1 & w_{43}^1 \end{bmatrix}^T \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$(4 \times 3) \rightarrow (3 \times 4)$ (4×1) $(3 \times 1) \rightarrow 3 \times 1$

$$= \begin{bmatrix} w_{11}^1 & w_{12}^1 & w_{13}^1 & w_{14}^1 \\ w_{21}^1 & w_{22}^1 & w_{23}^1 & w_{24}^1 \\ w_{31}^1 & w_{32}^1 & w_{33}^1 & w_{34}^1 \end{bmatrix} \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} w_{11}^1 x_{i1} + w_{12}^1 x_{i2} + w_{13}^1 x_{i3} + w_{14}^1 x_{i4} + b_{11} \\ w_{21}^1 x_{i1} + w_{22}^1 x_{i2} + w_{23}^1 x_{i3} + w_{24}^1 x_{i4} + b_{12} \\ w_{31}^1 x_{i1} + w_{32}^1 x_{i2} + w_{33}^1 x_{i3} + w_{34}^1 x_{i4} + b_{13} \end{bmatrix}$$

(3×1)

$$= \begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix}$$

Input to next layer

Layer # 2

$$\begin{bmatrix} w_{11}^2 & w_{12}^2 \\ w_{21}^2 & w_{22}^2 \\ w_{31}^2 & w_{32}^2 \end{bmatrix}^T \begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$(3 \times 2)^T \rightarrow (2 \times 3) \quad (3 \times 1) \quad (2 \times 1)$$

$$= \begin{bmatrix} w_{11}^2 & w_{12}^2 & w_{31}^2 \\ w_{11}^2 & w_{22}^2 & w_{32}^2 \end{bmatrix} \begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$= \begin{bmatrix} w_{11}^2 \cdot 0_{11} + w_{12}^2 \cdot 0_{12} + w_{31}^2 \cdot 0_{13} + b_{21} \\ w_{11}^2 \cdot 0_{11} + w_{22}^2 \cdot 0_{12} + w_{32}^2 \cdot 0_{13} + b_{22} \end{bmatrix}$$

$$= \begin{bmatrix} 0_{21} \\ 0_{22} \end{bmatrix} \rightarrow a^{(2)}$$

Input to next layer

Layer # 3

$$\begin{bmatrix} w_{11}^3 \\ w_{21}^3 \end{bmatrix}^T \begin{bmatrix} 0_{21} \\ 0_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$$(2 \times 1)^T \rightarrow (1 \times 2) \quad (2 \times 1) \quad (1 \times 1)$$

$$= \begin{bmatrix} w_{11}^3 & w_{21}^3 \end{bmatrix} \begin{bmatrix} 0_{21} \\ 0_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$$= \begin{bmatrix} w_{11}^3 \cdot 0_{21} + w_{21}^3 \cdot 0_{22} + b_{31} \end{bmatrix} = \begin{bmatrix} 0_{31} \end{bmatrix} = \hat{y}_i \rightarrow a^{(3)}$$

Input

So,

$$a^{(1)} = \sigma \left(\underbrace{2^{(0)}}_{\text{Input}} w^{(1)} + b^{(1)} \right)$$

$$a^{(2)} = \sigma \left[\underbrace{a^{(1)}}_{\text{Output of previous layer}} w^{(2)} + b^{(2)} \right]$$

$$a^{(3)} = \sigma \left[\underbrace{a^{(2)}}_{\text{Output of previous layer}} w^{(3)} + b^{(3)} \right]$$

✓

$$\sigma \left(\underbrace{\sigma \left(\underbrace{a^{(0)}}_{\text{Input}} w^{(1)} + b^{(1)} \right)}_{\text{Output of previous layer}} w^{(2)} + b^{(2)} \right) w^{(3)} + b^{(3)}$$

$$\underbrace{\hspace{10em}}_{\text{Output of previous layer}}$$

$$\underbrace{\hspace{10em}}_{\text{Output of previous layer}}$$